

How littlEye AI learns — and why it sometimes answers wrongly

This short guide explains how to read the model's results, why the model occasionally states nonsense with great confidence, and how to teach it with your own images, step by step, so that it improves exactly where you need it to.

1 · Why the model called normal lung “Heart 100 %”

A real-world experience: a web image of normal lung → the model said Heart 100 %; an image of grey hepatization in lobar pneumonia → Liver 81.8 %. This is not a random malfunction — there are three understandable reasons behind it:

- **Scale.** The baseline model littlEye-zaklad-v4 is trained on patches at roughly the 20× objective (field of view ~0.5 mm). An overview image of a whole lobe at low magnification is a completely unknown world to the model — alveoli appear as fine texture rather than the structures it knows.
- **The model MUST pick one of its classes.** “100 %” does not mean the certainty of a pathologist — it only means “of everything I know, this most resembles heart”. The model cannot say “I don't know” or “I have never seen this”.
- **The model knows only normal histology.** The Michigan collection contains no pathology. Grey hepatization — an airless lobe packed with cells — logically resembles liver to the model: dense, cellular, without alveolar spaces. The model has never seen pneumonia, so it picked the nearest thing it knows.

The golden rule

The model can only assess what it was trained on: the same scale (~20× objective; in practice a microphotograph at a total magnification of 100×–200×), the same stain (H&E;) and microphotograph quality — not a web JPEG. The more your image differs from the training images, the less the result means.

2 · How learning works — a dataset, not magic

A Core ML model cannot be fine-tuned — a finished model is baked. Learning in littlEye AI therefore works through the dataset: everything you want the model to learn is saved as a labelled image, and the model is retrained from scratch on the whole dataset. It takes minutes and has a great advantage: every new model comes with honest, measurable metrics on a held-out test.

3 · The learning loop — step by step

1. Evaluate an image (Classify image → From photos / From files).
2. Disagree with the result? Below the result there is a section “Learn from this image”: choose the correct organ and diagnosis and tap Save to dataset. The image is stored together with the correct answer.
3. In the organ's dataset, open the image detail and fill in the magnification and stain; if the image contains several findings, mark the location of the finding with a crop — the crop is then used for training.
4. Once you have collected at least 5 images per class this way (we recommend 30+), open Training and retrain the model. You may delete the older model of the same name.

5. Check the test % and the confusion matrix — you will see exactly what the model still confuses, and therefore which images to add next time.

This IS the learning

The loop evaluate → correct → save → retrain → verify is precisely how the app learns and remembers your corrections. With every iteration the model becomes a little more yours — and by sharing the dataset (.le3set) your corrections reach your colleagues as well.

4 · Editing diagnoses after training

- Add a new diagnosis: in the organ section tap Add next to “Diagnoses (training classes)” — the new class is immediately available for labelling and for the learning loop.
- Reassign an image: in the image detail simply choose a different diagnosis — the change takes effect at the next training run.
- Diagnoses bind to a model only through training. A finished model has its classes baked in — renaming or adding a diagnosis in the dataset changes only the next model you train.

5 · What to realistically expect from how many images

Images per class	What to expect
5–10	the model gets going, but treat the results as indicative only
30+	a usable working model on your own data
100+ from different cases	more stable behaviour also on slides from elsewhere
+ always	a “Normal” class, one magnification, one stain, validation on independent images

A note on the baseline model

littlEye-zaklad-v4.le3AI is a starter model of normal histology — it recognizes 20 tissues (held-out test 79.1 %) (Michigan, CC BY-NC-SA). It was built together with the dataset littlEye-zaklad-v2.le3set containing the 2,184 verified images it grew from. Its job is to recognize the tissue on a good-quality microphotograph at the right scale — not to diagnose pathology. Real diagnostic power comes only from models trained on your own slides through the learning loop above. Recommended approach: for every organ you actually assess, gradually fill in your own images (Normal + pathologies from your autopsies) and retrain.

Availability of the Michigan-derived models

The models derived from the Michigan collection (and from TCGA) are not being distributed at present — we are confirming with the University of Michigan Tech Transfer Office whether our reading of the CC BY-NC-SA 4.0 licence matches the rights holder's. The workflow described above works exactly the same with a model you train on your own slides.

Ján Šikuta, MD, PhD. · jansikuta@me.com · littlEye AI 2.2.2 · August 2026